

Fire Fighting Systems – Manual

Reference Manufacturer: Florian Security Systems S.A. (FSS)

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Florian Security Systems S.A. (FSS) is a Greek company specializing in fire protection equipment. Its Fire Fighting Systems range covers four main technologies: CO2 systems, Clean Agent (Inert Gas) systems, FIWAREC self-activating systems, and Fat & Oil systems for local application. This manual summarizes each of these systems, their principle of operation, applications, advantages, and the technical design tables used for CO2 system sizing.

1. About the Reference Manufacturer

Name: Florian Security Systems S.A. (FSS)

Country: Greece

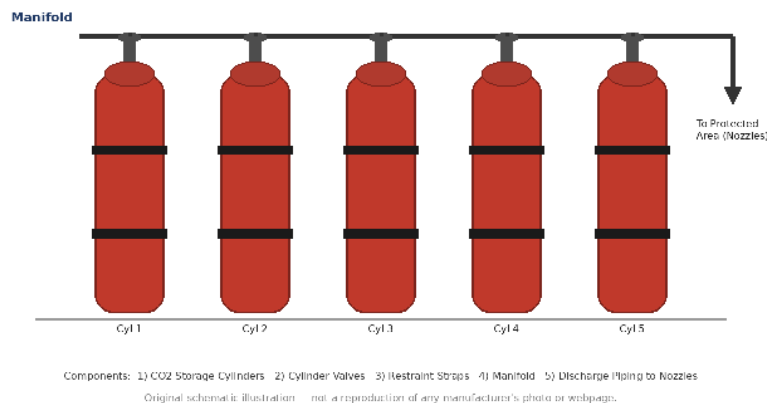
Over 25 years of experience developing CO2 fire extinguishing technology, and 15 years developing Clean Agent (IG541/IG01/IG100/IG55) systems.

Product line includes: portable and welded CO2 extinguishers, IG541 clean agent systems, FIWAREC self-activating systems, and topical application systems for fats & oils.

2. CO2 Fire Extinguishing Systems

F.S.S. has been making considerable advances in the development of CO2 technology regarding fire fighting for over 25 years, continuously updating and improving its fire fighting branch for hundreds of CO2 installations in refineries, industries, ships, handicrafts, electricity establishments, electric equipment, computers, etc.

CO2 Fire Extinguishing System – Cylinder Bank Layout



Original schematic illustration of a CO2 cylinder bank connected to a manifold and discharge piping.

2.1 System Components

High-pressure CO2 storage cylinders, arranged in a bank and secured with restraint straps.

A manifold that combines the individual cylinder lines into a single main feed line.

Manual/automatic valves fitted on each cylinder head.

Piping and nozzles that distribute the gas across the protected area.

Flexible hoses connecting each cylinder to the main manifold line.

2.2 Working Principle

A CO₂ system extinguishes fire by displacing oxygen from the fire area and reducing its concentration below the level needed to sustain combustion, along with a limited cooling effect from gas expansion during discharge. The gas is colorless, odorless, and electrically non-conductive, making it suitable for areas containing sensitive electrical and electronic equipment.

CO₂ is stored at room temperature in steel cylinders approved by the EEC. Cylinder pressure changes with temperature and is approximately 59 bar at 21°C. Multiple banks of cylinders can be arranged for concurrent release of CO₂ according to the plan's requirement, and delay-discharge nozzle systems and automatic prevention methods are available for crowded areas.

2.3 Typical Applications

Flammable liquid storage areas.

Flammable tanks / exhaust fume ducts.

Paint spray areas.

Electrical switchgear & substations.

Fryers / ovens.

2.4 Advantages of Carbon Dioxide

CO₂ is pure: a colourless, tasteless and inert gas. Once the fire is extinguished, it evaporates completely leaving no residue behind — nothing to clean, and unlike water there is no fear of damage. It is harmless to most materials, does not contaminate food, and is non-corrosive and non-conductive, making it safe on sensitive electric/electronic equipment and valuable works of art, paintings and manuscripts.

CO₂ is flexible: its three-dimensional action means it can fight fires vertically as well as horizontally. Its fast movement lets it penetrate obstacles through openings, reaching the most hidden spots.

CO₂ has a low cost: it is a stable commercial product with many other uses, accessible worldwide at low cost — essential for regular tank refilling in local application systems where fire danger is common.

Flexibility of design: F.S.S.'s wide range of systems allows machinery design appropriate to each customer's requirements. The system can be operated automatically or manually and adjusted to avoid hazards, with a large number of discharge nozzles. Automatic control can be mechanical, pneumatic, electronic, or a combination of these.

2.5 Design Tables for CO₂ Sizing

Table 1 — Concentrations for deep seated hazards (flooding factor):

Hazards	Flooding Factor
Dry electrical wiring and insulation	1.35 kg/m ³
Computer equipment	1.5 kg/m ³
Data processing and tape storage	2.25 kg/m ³
Record stores / archives	2.00 kg/m ³
Dust collectors	2.7 kg/m ³

Table 2 — Determine basic CO₂ quantity (by protected volume):

Volume - m ³	Factor - kg/m ³
>4	1.15

>4 – 14	1.07
>45 – 126	0.90

Table 3 – Determine higher concentrations for specific hazards (material conversion factor):

Material	Conversion Factor
Ethylene	1.6
Ethyl Ether	1.5
Acetylene	2.5
Benzol	1.1
Butadiene	1.3
Hexane	1.1
Kerosene	1.0
Petrol	1.0
Hydrogen	3.2

3. Clean Agent Fire Extinguishing Systems (IG541 / IG01 / IG100 / IG55)

F.S.S. has developed technology for these systems for 15 years, staying informed and continuously improving fire safety for hundreds of IG-541, IG-01, IG-100 and IG-55 installations.

Since the 1960s, protection of electronic equipment in occupied spaces was provided by HALON 1301 (a total-flooding system). It was clean and left no residue, but in 1987 it was revealed that HALON destroys the ozone layer (maximum Ozone Depleting Potential) and has a very high Global Warming Potential, contributing to the greenhouse effect. This made it essential to find replacement materials. The new systems – IG-541, IG-01, IG-100 and IG-55 – are Clean Agent Fire Extinguishing Systems that are as effective as HALON without its environmental drawbacks, and they leave no residue such as dust or water.

3.1 Formal Applications

Computer installations and places where files are stored.
 Installation of telecommunication, radio and television stations.
 Industrial installations and areas of flammable liquids.
 Laboratories.
 Banks.
 Places of cultural heritage (such as museums).
 Medical diagnostic centres.
 Engine rooms of boats.

3.2 Advantages of Clean Agent Fire Extinguishing Systems

Safe for the environment: these systems use inert gases (N₂, Ar and CO₂) naturally present in atmospheric air, so they have null ODP and GWP. After fire fighting they fully vaporize, leaving no residue and not affecting surfaces or contaminating food. Any leaked gas keeps its initial density in the protected space, and the reduced-oxygen mixture created still allows a person to breathe.

Inert gases are flexible: their three-dimensional action extinguishes fires vertically and horizontally, and they can penetrate obstacles to reach concealed points.

Approvals: listed in ISO 14250, NFPA 2001, CEA 4008, EPA and VdS for extinguishing fires of categories A, B, C, and fires in the presence of electric current.

Flexibility of planning: as F.S.S. produces a wide spectrum of systems, the extinguishing mechanism can be tailored to each customer's requirements, including retrofitting existing extinguishing lines. IG-541, IG-01, IG-100 and IG-55 are stored in high-pressure cylinders (200 or 300 bar) meeting the latest European specifications (97/23/EC with CE marking or 99/36/EC with π marking). Valves meet EN 849 or 97/23/EC standards and carry CE marking and a manometer indicating stored pressure at all times.

4. FIWAREC Self-Activating Systems

FIWAREC systems are self-activating fire suppression systems built around a pressurized sensor tube that acts as a linear heat and flame detector. The tube is flexible and tough enough to withstand harsh conditions, and can be routed throughout an installation to follow the hazard area. Upon flame or heat impingement, the tube ruptures at the hottest point and, by depressurization, activates the cylinder valve – no external energy or electrical power is needed for detection or actuation.

4.1 Direct Systems

Direct Systems are efficient, simple, self-activating systems for small enclosed hazard areas, and are failsafe against malfunction. Upon flame impingement or heat, the pressurized sensor tube connected to a cylinder containing the suppression agent ruptures at the hottest point, and the agent is discharged directly through the burst hole to the heart of the fire. Flexible and fast acting.

4.2 Indirect Systems

Indirect Systems are efficient, self-activating systems suitable for larger enclosed and even open hazard areas. Upon flame impingement or heat, the pressurized sensor tube bursts and, by depressurization, activates the cylinder valve to open. The agent is then released through separate pipe work and nozzles aimed at the hazard area. Flexible and fast acting.

4.3 Outstanding Features

Linear heat and flame detection allows for an unlimited number of detection points.

Flexible sensor tubing follows hazard and risk areas – even difficult-to-reach areas – resulting in fast reaction.

Failsafe by design principles and simplicity.

Flexible tube remains unaffected by dirt, dust, debris, oil and vibrations in nearly all industrial environments.

No need for any external energy or power supply for detection or actuation.

4.4 Applications

Transport / Leisure: buses, boats, trains – engine compartments.

CNC machines (automated operations): injection moulding machines, robotic welding machines, machining centers.

Electrical cabinets and controls: electrical distribution cabinets, process control cabinets, communication racks.

Industrial equipment / construction: fork lifts, machinery engine protection, industrial process equipment, generators, air compressors.

Local application systems (main, most important application): kitchens, roaster-baker and fast-food stores.

4.5 Range of Extinguishants

Rühl Favorit Home (foam for oil & grease).

AFFF (foam).

ABC powder.

CO2.

FM200.

Water.

Inert gases (IG541 - IG01 - IG100 - IG55).

5. Fat & Oil Systems for Local Application

Complex and unique requirements for perfect fire protection in local applications do not necessarily need complicated product arrangements. Simplicity, best performance and common sense are F.S.S.'s answers to reaching the highest efficiency and safety. In many situations, extinguishing a fire at the source minimizes damage to valuable assets and enables a quicker recovery of operations.

F.S.S.'s automatic fire suppression systems build on years of experience and technical knowledge, and can be tailored to customer requirements while sharing common features:

Simple to install and cost effective.

Totally self-contained, requiring no electrical source, and remaining operational during power interruption.

Systems include detection/discharge tubing, specialist valves/regulators, and a pressure vessel containing an appropriately selected fire extinguishing agent.

5.1 Principles of Operation

The heart of an F.S.S. Automatic Fire Suppression System is its special sensor tubing, which acts as a linear heat and flame detector. The pressurized sensor tubing is completely flexible and tough enough to withstand harsh conditions, and can be run throughout the installation to follow the hazard area. Upon flame or heat impingement, the tube ruptures at the hottest point and, by depressurization, the cylinder valve is activated to operate the extinguishing system. No external energy or electrical power is needed for detection or actuation.

Linear heat and flame detection allows for an unlimited number of detection points.

Flexible sensor tubing follows hazard and risk areas, even difficult-to-reach areas, resulting in fast reaction.

Failsafe by design principles and simplicity.

Flexible tube remains unaffected by dirt, dust, debris, oil and vibrations in nearly all industrial environments.

No need for any external energy or power supply for detection or actuation.

6. Safety Precautions (CO2 Systems)

CO2 is lethal at fire-extinguishing concentrations due to oxygen displacement; the area must be evacuated immediately upon hearing the alarm, before discharge.

Audible and visual alarm systems and a time delay must be provided before automatic discharge to allow evacuation.

Manual release controls are installed at the cylinders, with a remote control near the main

exit inside a break-glass protective enclosure.

Re-entry into the protected area after discharge is prohibited until oxygen levels are confirmed to have returned to normal.

CO2 cylinders must be inspected periodically (weight, pressure, and hydrostatic test validity) to ensure none are past their inspection due date.

7. Maintenance and Periodic Inspection

Periodically check cylinder weight to confirm there is no gas leakage.

Inspect valves, flexible hoses, and the manifold for leaks or damage.

Periodically test free air flow through the piping and nozzles.

Confirm cylinders are securely fastened and not inclined beyond the allowed angle.

Document all inspection and maintenance activities in the system maintenance log.

Reference: Florian Security Systems S.A. (FSS) — www.fss.gr, in addition to general technical references on CO2, clean agent, and self-activating fire extinguishing systems. The diagram in Section 2 is an original schematic created for this manual, not a reproduction of any manufacturer's photo or webpage.